



Base Metals One Year Forecast

January 2011

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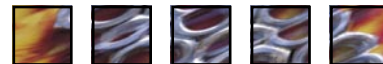
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Introduction

Base metals finished 2010 on a firm note with exceptional gains seen in December, ranging from 9.6% for aluminium to as high as 17.1% for lead. GFMS' Base Metals Index, which gives equal weight to all the base metals unlike the LME index that is heavily skewed towards aluminium and copper, gained 14.2% during December. The index started the year at 292.75, troughed at 231.84 on June 8 and then ended the year on a high of 363.4. The extreme nature of the rally in 2010 left the index only just below the all-time high achieved in the previous bull market of 394.98. On January 6 2011, the GFMS Base Metal index stood at 365.42. In terms of the relative performance of average annual prices, the percentage gains achieved last year were as follows:

Al 30.5; Cu 46.0; Ni 48.4; Pb 24.0; Sn 50.4; Zn 30.1

Although the percentage increases are obviously also a function of the price performance in 2009 which sets the base, the two top performing metals copper and tin have in our view the best fundamentals. We expect this to continue to be the case in 2011. The experience of aluminium and zinc merely confirms the fact that in the current environment, when there is strong investment activity an improvement in the fundamentals is not a "necessary" condition for price increases. The concept of relative performance of individual metals is becoming increasingly important. This is one of the reasons we could see further advances for aluminium despite the stock overhang.

Many of the broader drivers of the market last year were positive – weak dollar, investment fund activity, China, accelerating global economic growth, relatively low downstream inventories and continuing supply shortfalls. We will consider which factors will continue to support the market in 2011 below. Although our latest supply-demand projections are similar to those released in December, GFMS have amended our price projections for 2011. With the exception of tin, this has involved an increase to our projections (see individual sections for our price forecasts).

What will be the key drivers in 2011?

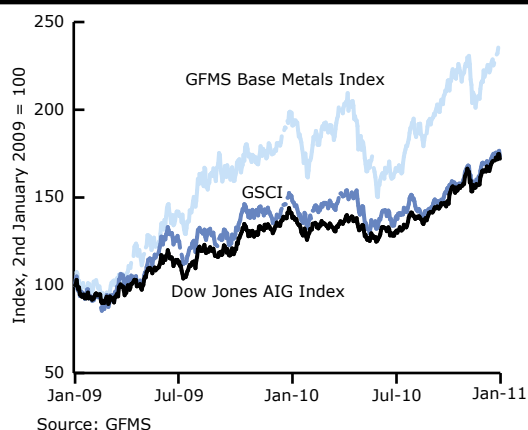
There was an exceptionally strong rebound in demand across all metals in 2010. The most notably feature was the surge in offtake in the mature economies, which exceeded that of China for the first time in a number of years! The data on crude steel production, which given its myriad of end uses is a reasonable proxy for industrial commodity demand, reinforces this point. Chinese crude output increased by 10.1% last year, while output outside of the country increased by 21.6%.

Chinese base metal consumption growth was held back by destocking; however as we have noted in our earlier reports underlying non-ferrous semis activity and steel output was strong in the country. For example stainless and galvanized steel output rose by 25% and 60% respectively. Imports of some base metals into China slipped earlier in the year. However, more recently, they have started to show month-on-month-gains.

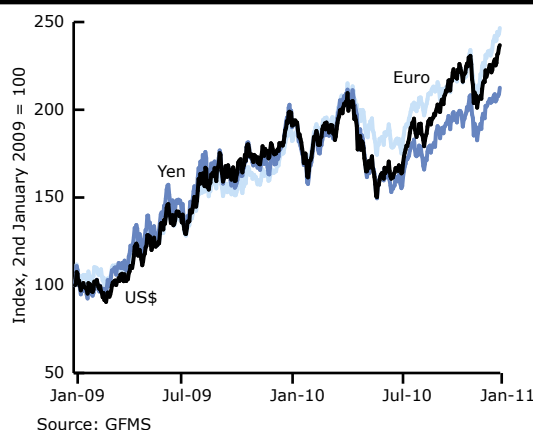
In terms of Chinese production, the market focused on the reduction in supply to power-intensive users such as the base metals sector as part of an attempt to fulfill energy commitments in the most recent Five Year plan. There are also broader measures to curb small-scale inefficient smelting and refining capacity. However, analysis of the latest monthly production data shows that these measures have been primarily focused on aluminium with production in November 16% below the peak for the year seen in the summer. For lead, for example, production in November was at an all-time high, while zinc output was just below the October peak. Chinese raw material production – alumina and concentrate output increased sharply in 2010, up to 41% and 22% respectively in the case of lead and zinc. This enabled the gains in refined production despite lower imports of raw materials.

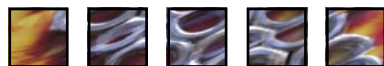
For the mature economies, base metal demand will not benefit from the one-off restocking rebound seen in 2010.

Base Metals & Other Commodity Prices



Base Metals Prices in Major Currencies





Base Metals Supply-Demand Overview

(000 tonnes)	2004	2005	2006	2007	2008	2009	2010
Aluminium							
Global Production	29,922	32,017	33,975	38,056	39,479	36,922	41,031
Global Consumption	30,285	32,040	34,275	37,941	37,375	35,007	40,315
Metal Balance	-363	-24	-300	116	2,104	1,915	716
LME Cash (\$/tonne)	1,717	1,898	2,567	2,645	2,571	1,668	2,173
Copper							
Global Production	15,928	16,572	17,291	17,934	18,200	18,367	18,929
Global Consumption	16,803	16,601	17,139	18,152	17,893	17,471	18,997
Metal Balance	-875	-29	152	-218	307	896	-68
LME cash (\$/tonne)	2,868	3,864	6,731	7,126	6,952	5,164	7,539
Lead							
Global Production	6,980	7,626	7,933	8,121	8,650	8,709	9,295
Global Consumption	7,295	7,783	8,061	8,176	8,649	8,630	9,198
Metal Balance	-315	-157	-128	-55	1	79	97
LME Cash (\$/tonne)	888	976	1,288	2,600	2,085	1,726	2,148
Nickel							
Global Production	1,251	1,288	1,336	1,388	1,353	1,327	1,429
Global Consumption	1,251	1,296	1,366	1,356	1,294	1,299	1,466
Metal Balance	0	-8	-30	32	59	28	-37
LME Cash (\$/tonne)	13,850	14,733	24,287	37,181	21,029	14,700	21,812
Tin							
Global Production	343	350	351	350	343	333	347
Global Consumption	334	345	363	357	355	322	366
Metal Balance	9	4	-12	-7	-12	11	-19
LME Cash (\$/tonne)	8,513	7,370	8,763	14,580	18,499	13,593	20,447
Zinc							
Global Production	10,395	10,220	10,654	11,359	11,646	11,267	12,391
Global Consumption	10,646	10,612	11,014	11,307	11,437	10,851	12,111
Metal Balance	-251	-392	-360	52	209	416	280
LME Cash (\$/tonne)	1,048	1,382	3,273	3,250	1,870	1,648	2,159

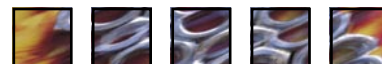
Source: GFMS, Various

Also the move towards fiscal tightening and the ending of specific measures such as the auto scrapping will affect demand growth. On a more positive note for base metal producers downstream inventories are fairly tight, also the anecdotal evidence that we receive suggest that demand is holding up well at present. Nevertheless after the sharp recovery seen in 2010, we are not building further significant advances into our supply demand balance calculations.

Although we have noted on numerous occasions that it is difficult to double guess the funds, many of the factors that attracted to the base metal sector are still in place. Base metals as an asset class continue to gain acceptance. This year the market will benefit from the launch of various ETFs. While success of these instruments is yet to become clear, some tonnages will be removed from the market.

The base metal sector is increasingly being seen as part of the diversification of asset allocation. In line with a number of other markets, particularly equities, investor interest remains high in a low interest rate environment especially when the bulk of the economic data remains positive.

We have seen the downside potential during 2010 during periods of risk aversion, when the sovereign debt problems and some concerns over Chinese growth helped to trigger a massive fall in the base metal markets. GFMS' Base Metals index fell from 316.44 on April 26 to a low for the year of 231.84 on June 8. There is obviously the potential for a repeat of this with the debt issue in Europe not resolved, but merely shifted to other countries within the region. Also the massive US deficit has not been tackled, while the potential for growth to stall in China remains in the background.



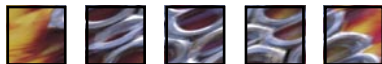
LME Prices Overview

	Prices US/tonne						% change year-on-year					
	Al	Cu	Ni	Pb	Sn	Zn	Al	Cu	Ni	Pb	Sn	Zn
Monthly												
Nov-09	1,949	6,676	16,991	2,309	14,942	2,193	5.2	79.6	58.8	78.8	9.5	90.3
Dec-09	2,180	6,982	17,066	2,329	15,547	2,376	46.3	127.3	76.2	141.8	38.3	115.9
Jan-10	2,235	7,386	18,439	2,368	17,715	2,434	58.2	129.3	63.1	109.1	55.8	105.0
Feb-10	2,049	6,848	18,976	2,124	16,362	2,157	54.0	106.6	82.3	93.0	48.2	94.0
Mar-10	2,206	7,463	22,461	2,172	17,549	2,275	65.1	99.0	131.6	75.3	64.4	87.0
Apr-10	2,317	7,745	26,031	2,265	18,684	2,367	63.1	75.8	133.1	63.8	59.1	71.6
May-10	2,041	6,838	22,008	1,883	17,566	1,968	39.7	49.7	74.2	30.7	27.4	32.7
Jun-10	1,931	6,499	19,389	1,704	17,320	1,743	22.7	29.6	29.6	1.8	15.6	11.9
Jul-10	1,988	6,735	19,518	1,837	18,191	1,844	19.2	29.1	22.1	9.4	29.6	16.8
Aug-10	2,118	7,284	21,413	2,075	20,755	2,045	9.5	18.1	9.0	9.2	39.6	12.2
Sep-10	2,162	7,709	22,643	2,184	22,701	2,151	17.9	24.4	29.6	-0.9	52.7	14.2
Oct-10	2,347	8,292	23,807	2,380	26,343	2,372	24.9	31.9	28.5	6.2	75.5	14.5
Nov-10	2,333	8,470	22,909	2,377	25,519	2,292	19.7	26.9	34.8	2.9	70.8	4.5
Dec-10	2,351	9,147	24,111	2,413	26,163	2,281	7.8	31.0	41.3	3.6	68.3	-4.0
Quarterly												
2009 Q2	1,488	4,676	12,992	1,506	13,551	1,476	-49.4	-44.6	-49.5	-35.0	-40.1	-30.2
2009 Q3	1,806	5,840	17,614	1,925	14,576	1,757	-35.3	-24.1	-7.2	0.5	-29.0	-0.9
2009 Q4	2,001	6,643	17,543	2,292	15,164	2,211	9.3	68.6	61.2	83.2	15.5	86.0
2010 Q1	2,165	7,243	20,078	2,219	17,225	2,288	59.2	110.9	92.0	91.4	56.3	94.9
2010 Q2	2,092	7,013	22,382	1,944	17,844	2,018	40.5	50.0	72.3	29.1	31.7	36.7
2010 Q3	2,089	7,242	21,188	2,031	20,546	2,013	15.7	24.0	20.3	5.5	41.0	14.6
2010 Q4	2,343	8,634	23,598	2,390	26,001	2,315	17.1	30.0	34.5	4.3	71.5	4.7
Annual												
2005	1,898	3,684	14,733	976	7,370	1,382	10.6	28.4	6.4	9.8	-13.4	31.9
2006	2,567	6,731	24,287	1,288	8,763	3,273	35.2	82.7	64.9	32.0	18.9	136.8
2007	2,639	7,126	37,181	2,595	14,536	3,250	2.8	5.9	53.1	101.5	65.9	-0.7
2008	2,571	6,952	21,029	2,085	18,499	1,870	-2.6	-2.5	-43.4	-19.7	27.3	-42.5
2009	1,668	5,164	14,700	1,726	13,593	1,659	-35.2	-25.7	-30.1	-17.2	-26.5	-11.3
2010	2,173	7,539	21,809	2,148	20,447	2,159	30.3	46.0	48.4	24.4	50.4	30.1

Source: GFMS, LME

In terms of our price forecasts, we would note that the corrections in base metals even post Lehman, while extreme, tend to be short-lived. While we are expecting periods of weakness and heightened volatility given the high base level, we are generally forecasting high period averages in 2011. The average annual price for all the base metals should be higher in 2011 than in 2010 (see individual sections for

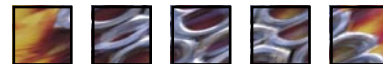
details), and the peak price in this year should be higher than last. However, compared to where prices reside in early 2011, the main theme of 2011 should be consolidating some of the extraordinary gains seen in late 2010.



LME Inventory Overview

	End-Period (000 tonnes)						Number of weeks' W. World Consumption					
	Al	Cu	Ni	Pb	Sn	Zn	Al	Cu	Ni	Pb	Sn	Zn
Monthly												
Nov-09	4,600	441	141	138	27	455	12.2	2.3	10.1	1.6	8.3	4.0
Dec-09	4,624	502	158	147	27	489	12.3	2.6	11.4	1.6	8.3	4.3
Jan-10	4,611	544	166	158	28	497	10.8	2.6	10.7	1.7	7.6	4.0
Feb-10	4,570	551	163	167	25	541	10.7	2.6	10.5	1.8	6.8	4.4
Mar-10	4,594	512	158	176	24	542	10.7	2.4	10.1	1.9	6.6	4.4
Apr-10	4,530	497	146	182	21	559	10.6	2.3	9.4	2.0	5.8	4.5
May-10	4,556	476	138	193	20	620	10.6	2.2	8.9	2.1	5.6	5.0
Jun-10	4,422	449	124	190	17	617	10.3	2.1	8.0	2.0	4.8	5.0
Jul-10	4,381	413	118	184	15	620	10.2	1.9	7.6	2.0	4.1	5.0
Aug-10	4,435	399	120	191	14	621	10.3	1.9	7.7	2.0	3.9	5.0
Sep-10	4,352	374	124	194	12	616	10.2	1.8	7.9	2.1	3.4	5.0
Oct-10	4,306	369	128	200	13	622	10.0	1.7	8.2	2.1	3.6	5.0
Nov-10	4,289	355	132	204	15	631	10.0	1.7	8.5	2.2	4.2	5.1
Dec-10	4,275	378	137	209	16	701	10.0	1.8	8.8	2.2	4.5	5.7
Annual												
2005	644	92	36	44	17	394	1.5	0.4	1.7	0.4	3.6	2.7
2006	698	191	7	41	13	90	1.6	0.8	0.3	0.4	2.8	0.6
2007	929	199	48	45	12	88	2.1	0.9	2.5	0.5	3.0	0.6
2008	2,338	341	79	45	8	253	5.1	1.5	4.3	0.5	2.0	1.9
2009	2,811	491	84	54	9	345	7.5	2.5	6.0	0.6	2.8	3.0
2010	3,227	537	99	60	9	358	8.6	2.7	7.1	0.7	2.7	3.1

Source: GFMS, LME



Economic Indicators

Although obviously risks remain, we believe that the bulk of the economic data released since the December issue of the Base Metals One Year Forecast is broadly supportive to underlying base metal demand. Certainly the positive data contributed to the strong price performance in December.

In the middle of December, the **Chinese** data was strong. Chinese industrial production expanded strongly in November – up 13.3% year-on-year after most had expected further slowing. Urban Fixed Asset investment was also higher-than-expected – up 24.9% year-to-date – and ending the steady decline since the beginning of the year. Given the high correlation between this and base metal demand, this is

very positive for the sector. However, this has led to further fears on asset inflation and may lead to further efforts to cool the sector.

Elsewhere in Asia, **Taiwanese** industrial output was up 19.4% year-on-year as exports and domestic demand continued to perform well. Indian industrial production was up 10.8% year-on-year in October – a bounce back from September's disappointment.

Japanese output rose 1% month-on-month in November and was up 4.2% year-on-year. Nevertheless, with the end of inventory re-stocking, output growth is moderating and

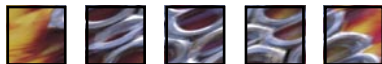
Main Economic Indicators

	Mar-10	Apr-10	May-10	Jun-10	Jul-10	Aug-10	Sep-10	Oct-10	Nov-10	Dec-10
Manufacturing PMI*										
Eurozone	56.6	57.6	55.8	55.6	56.7	55.1	53.7	54.6	55.3	56.8
USA	59.6	60.4	59.7	56.2	55.5	56.3	54.4	56.9	56.6	57.0
China	57.0	55.2	52.7	50.4	49.4	51.9	52.9	54.8	55.3	54.4
Japan	52.4	53.8	54.7	53.9	52.8	50.1	49.5	47.2	47.3	48.3
OECD Composite Leading Indicators**										
OECD	102.5	102.6	102.7	102.6	102.5	102.5	102.5	102.6		
Euro zone	103.4	103.6	103.6	103.6	103.6	103.4	103.4	103.4		
France	103.9	103.4	102.9	102.4	102.1	102.1	102.1	102.2		
Germany	103.7	104.5	105.1	105.5	105.7	105.3	105.2	105.2		
UK	103.8	103.6	103.3	103.0	102.7	102.4	102.3	102.3		
USA	102.0	102.4	102.4	102.3	102.3	101.9	102.0	102.3		
NAFTA	102.3	102.5	102.5	102.4	102.3	101.9	102.0	102.2		
Japan	102.0	102.3	102.5	102.6	102.8	102.7	102.7	102.6		
Brazil	100.9	100.8	100.5	100.1	99.5	99.5	99.1	98.9		
China	104.0	103.6	103.0	102.0	100.9	100.9	101.3	101.9		
India	101.4	101.2	101.1	100.9	100.7	100.9	100.8	100.6		
Russia	100.7	101.4	102.1	102.8	103.5	104.0	104.8	105.5		
Industrial Production (m-o-m)										
Euro-zone	1.8	0.6	1.1	-0.2	0.1	1.1	-0.9			
Germany	3.0	1.0	3.4	-0.6	0.1	1.6	-0.8			
France	0.9	-0.5	1.8	-1.6	0.9	0.1	0.0			
Italy	0.1	1.0	0.9	0.5	0.3	1.6	-2.1			
UK	1.9	-0.7	0.6	-0.4	0.3	0.3	0.3			
USA	0.6	0.5	1.2	0.0	0.8	0.2	-0.2	0.0		
Japan	1.2	1.3	0.1	-1.1	-0.2	-0.5	-1.6	-1.8		
South Korea (y-o-y)	22.7	20.1	21.7	17.2	15.6	17.0	3.9	13.5		
Brazil	3.4	-0.9	-0.2	1.2	0.5	-0.2	-0.2			
China (y-o-y)	18.1	17.8	16.5	13.7	13.4	13.9	13.3	13.1	13.3	
India	12.9	-13.3	0.0	0.6	6.2	-6.6	0.8	1.8		
Consumer/Business Confidence indicators										
Eurozone Economic Sentiment	97.9	100.6	98.4	98.9	101.1	102.3	103.3	103.8	105.3	
USA Consumer Confidence	52.3	57.7	62.7	54.3	51.0	53.2	48.6	49.9	54.3	52.5
Japan Consumer Confidence	41.0	42.1	42.7	43.6	43.4	42.5	42.4	41.1	40.6	
Brazil Consumer Confidence	112.3	113.0	115.3	117.0	117.4	120.4	122.5	123.1	126.2	124.2
China Consumer Confidence	107.9	106.6	108.0	108.5	107.8	107.3	104.1	103.8	102.9	

* PMI: 50=no change m-o-m, >50 = expansion, <50 = contraction

** OECD CLI: CLI is increasing and >100 = the economy is in the expansion phase of the business cycle, CLI is decreasing but >100 = the economy is in a downturn, CLI is decreasing and <100 = the economy is in a slowdown, CLI is increasing but <100 = the economy is in recovery.

Source: OECD, National Statistics, Dismal Scientist & GFMS



the strong yen continues to put downward pressure on exporters.

The latest ISM manufacturing index in the **United States** rose from November's 56.6 to 57.0. This marks the second increase in the past three months and puts the index at its highest since May. A particularly encouraging sign going forward was the 4.3 point increase to the new orders component of the index to 60.9. Overall, the latest index is consistent with manufacturing activity on an upward trend in early 2011.

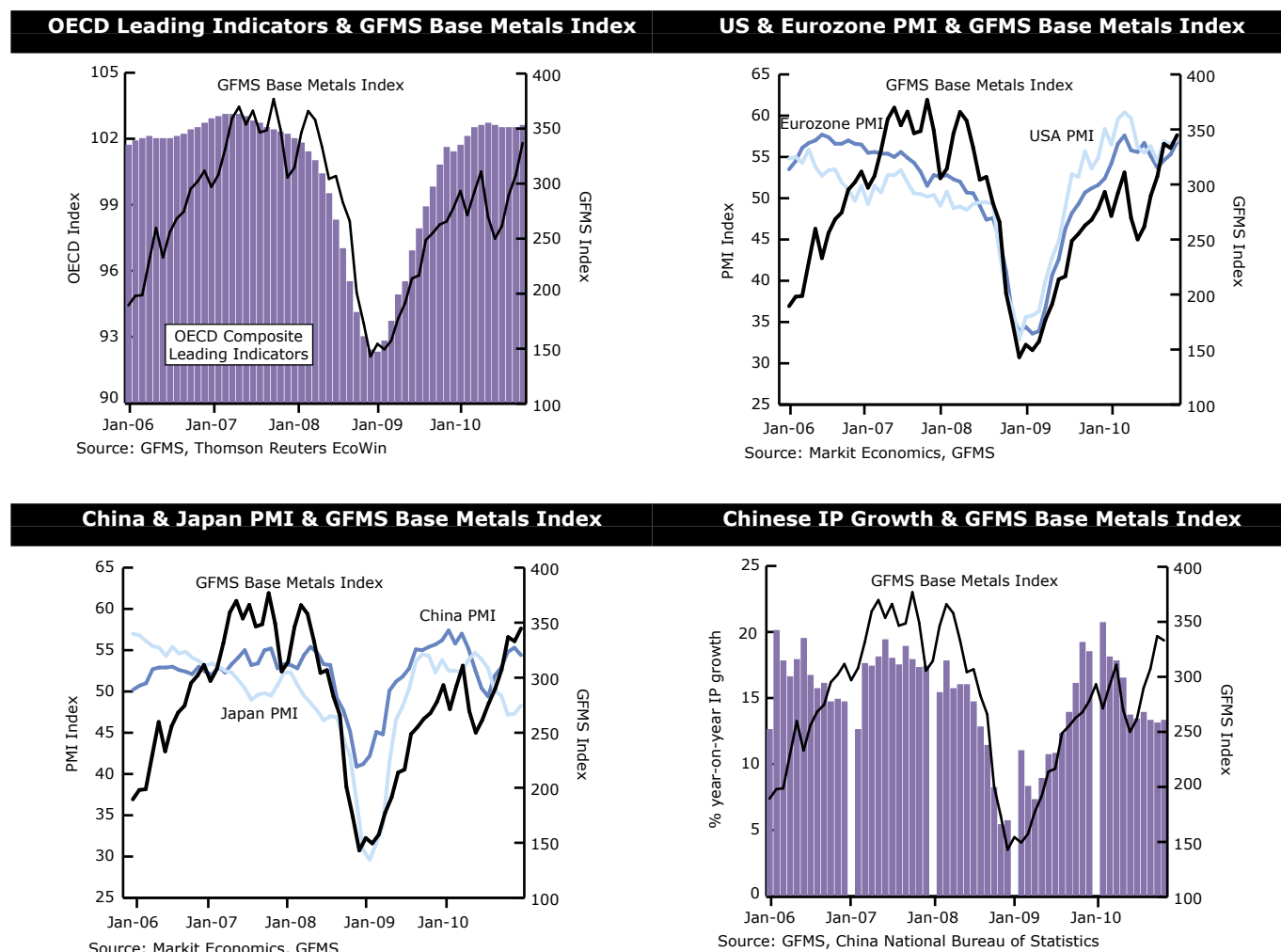
Consumer sentiment improved in December according to the University of Michigan. Its index rose to 74.5, up from 71.6 in November. While the recent gains in confidence are welcome, there are still questionmarks overhanging the consumer sector notably the weak job environment and higher energy prices.

The US auto industry finished 2010 on a high with sales at a seasonally adjusted annualised rate of 12.55 million units. Inevitably discounting has played its role. However underlying demand is improving given low sales earlier in the year. This pent up demand has yet to be sated and low

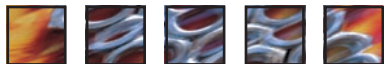
interest rates and a Federal "tax break" should maintain some upward momentum despite still high unemployment

For **Europe**, our favoured indicator is the euro zone economic sentiment index. This increased to 106.2 in December from 105.1 in November and is at its highest level in over 3 years. Better activity in Germany (see below) was a key driver. Going forward the region will have to cope with the negative impact of fiscal tightening and still high unemployment. Nevertheless growth in the region has consistently surprised to the upside, albeit from a low base.

Manufacturing orders in Germany increased by 5.2% month-on-month on a seasonally adjusted basis in November. The strong performance was supported by a sharp rise in exports. Domestic orders also grew in December but at a much slower pace than external demand. Consumption and business spending are being boosted by rising confidence and growth, which has been stronger than expected. Underlying trends in German manufacturing remain encouraging for the coming months. The manufacturing PMI remains well above the neutral level of 50 and reached a five-month high in December.



Note: The GFMS Base Metals Index is an average of the six base metals indexed prices with equal weights. This is in contrast to the LME index, which is heavily weighted towards aluminium and copper.



Copper

Recent developments

- *Three-month futures prices rose to a record high of nearly \$9,900/tonne on 3rd December in New York. Supply concerns were a key driver of the rise, as the main port through which Collahuasi ships material remains closed.*
- *No more than two weeks following the resolution of what had been the longest ever strike at a major private mine in Chile, an accident at Collahuasi's main port saw the facility close. Xstrata has recently commenced shipping material from Antofagasta's port but continues to operate under force majeure.*
- *Reported inventories have been rising, most probably due to Chinese unreported stockpiles having been in large part returned to the market over the last couple months of the year.*
- *During December and indeed through to the present time, a single institution's holdings accounted for more than 50% (at peak close to 90%) of total LME stocks. The identity of the institution and the purpose of this dominant position has not been confirmed.*
- *The first ever physically backed copper ETF started trading in London in December. To-date inflows have been modest and far short of what market participants had feared.*

Outlook for the next 12 months

- *The recent price-hike has in our view both pushed the bottom of the range upwards and set a precedent for prices trading in the upper \$9,000s. Prompted by this development, GFMS have revised upwards our Base Case projections for the year and now forecast prices to average at \$9,000/tonne overall.*
- *Aside from shifting our projections upwards, we maintain our view that prices are due a notable correction from current levels and that they will later recover to comfortably exceed the recent high before the end of the year.*
- *As far as the supply-demand fundamentals are concerned, we continue to expect healthy increases in consumption against a backdrop of structurally constrained supply.*

GFMS' Forecast Scenarios

Base Case, 50% Probability

Represents what GFMS consider the most likely outcome for the markets.

Scenario B, 30% Probability

Faster recovery than under our Base Case in the near-term and stronger growth thereafter.

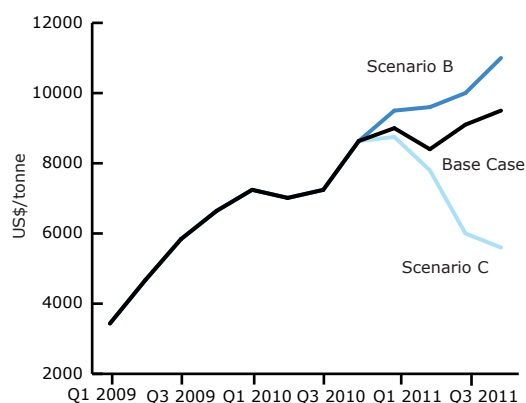
Scenario C, 20% Probability

The recovery that has been seen in the mature economies thus far slows markedly while growth in China continues to moderate as the impact of the stimulus packages wear off.

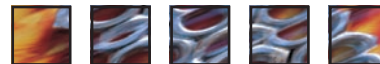
Copper Supply-Demand & Price Forecast

	2011			
(000 tonnes)	Q1	Q2	Q3	Q4
Supply/Demand				
Forecast (Base Case)				
Production	4,730	4,890	4,955	5,027
Consumption	4,700	4,870	5,050	5,114
Balance	30	20	-95	-87
Stocks	1,296	1,316	1,221	1,134
Price Forecast (\$/tonne)				
Base Case	9,000	8,400	9,100	9,500
Scenario B	9,500	9,600	10,000	11,000
Scenario C	8,750	7,800	6,000	5,600

Quarterly Copper Price & Forecast



Source: GFMS



Market background

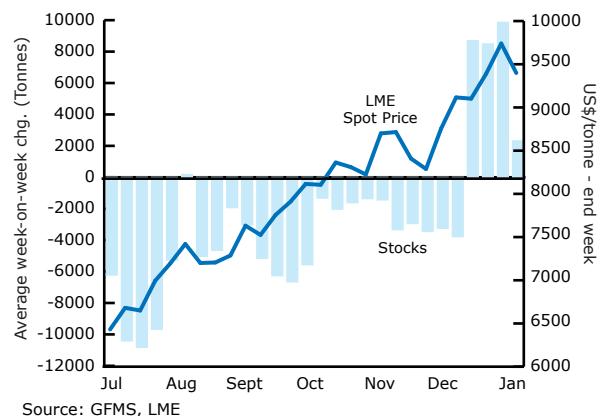
Copper had an upbeat start to the New Year, as the three-month futures price peaked at a little less than \$9,900/tonne on an intra-day basis in New York on 3rd January, while the LME was closed. The peak followed a strong performance throughout the month of December, on the back of expectations for ongoing growth in consumption coupled with persistent concerns for supply tightness fuelling investment demand for the metal.

Regarding the former, the announcement early in the month that Chinese authorities would refrain from increasing interest rates and instead opt for a "softer" tightening of monetary policy, involving higher reserve requirements, fuelled speculation that 2011 will be yet another year of strong growth for copper demand. This was compounded by November trade data being released, showing stronger than expected imports of the "red metal" into the country.

Yet more important in adding fuel to the copper price rally than this optimism towards Chinese consumption were developments in copper mining during the month. Indeed, the catalyst that saw prices move from the lower to the upper-\$9,000s was the accident at the main sea port terminal serving Xstrata's Collahuasi mine in Chile, discussed in detail later in this section.

It would of course be erroneous to ignore the impact of the ongoing inflows into the wider commodity sector during the month on copper investment and, by implication, prices. The sector (and indeed most pro-cyclical asset classes) has continued to benefit from the loose monetary policy environment, maintained in most major markets. Interestingly, the impact of persistently troubled sovereign debt markets on investor sentiment was limited during the month and focused around the time when Moody's downgraded Ireland's credit rating. Comments by the head of the IMF around the same time, suggesting Spain is likely to avoid need of a rescue and expressing confidence that the existence of the euro is not under threat, may have eased concerns somewhat.

Weekly Change in LME stocks vs Spot Price

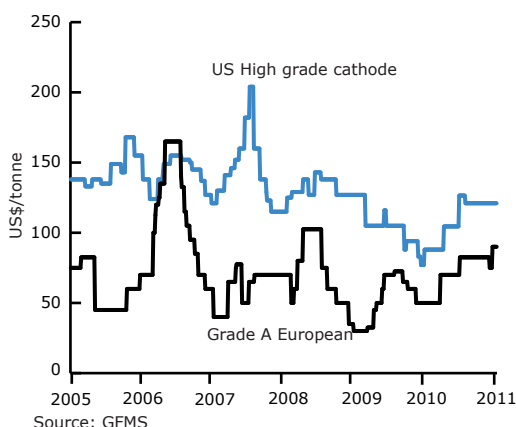


Our general thesis for the twelve months to follow remains broadly unchanged from the previous edition of this report, although the far more dramatic end-of-year rally than we had expected has led us to revise our price projections upwards for all four quarters. We continue to view the market as overbought and feel that a correction that is likely to affect most asset classes is due. The trigger for such a move could come in the form of renewed concerns towards the solvency of certain economies or a notable deterioration in the outlook for China. "After the dust settles", we continue to see prices resuming their uptrend, boosted by continuously improving fundamentals and what seems to be an insatiable appetite for copper by investors.

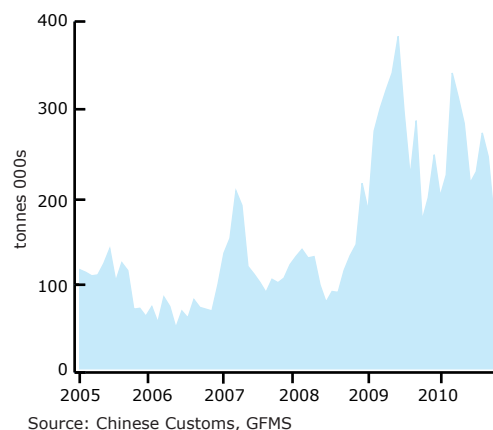
Inventories rise, as Chinese unreported stocks are released back into the market

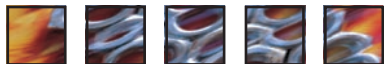
December saw the first monthly increase in LME inventories since February, amounting to just under 23,000 tonnes and a further 1,850 tonnes were added to stocks in the first few days of the New Year. Despite this fact, stocks remain low and amounting to just over one week's consumption (based on GFMS' 2010 estimate). SHFE stocks also rose over the month, by a little more than 9,000 tonnes, to reach approximately 132,000 tonnes. As discussed later on, GFMS believe that the increases in reported inventories are likely to reflect material that was held in unreported stockpiles in China being released back into the market.

Copper Premiums Jan 05 - Present



Chinese Copper Imports





Dominant position accounts for 90% of LME stocks

A recurring theme in the copper market throughout December was the exceptionally high portion of LME stocks held by a single dominant position. This stood above 50% for much of the month and reportedly peaked at close to 90% just before the Christmas holiday. Although there was much speculation about both the institution behind the position and its aim, to-date there has been no official confirmation for either.

The LME has strict guidelines forcing the holder of any amount in excess of 50% of total stocks to lend the material back to the market at a specified rate, to prevent the market from being "squeezed". Nevertheless, the very news of the situation developing is likely to have boosted both prices and the backwardation, and it is worth noting that the cash-to-three-months peaked at \$75/tonne on the last trading day of the year.

Chinese unreported inventories are released back into the market

The Chinese physical market seems to have been more than adequately supplied throughout December. This was due to much of the unreported inventory build that took place in 2009 being released into the market. Our recent research trip to the country confirmed our suspicions that such stocks have been declining in recent months and it is likely that this accelerated in November and December, partly triggered by rising prices and partly by some holders' need of the liquidity. This resulted in spot premia in the country coming under pressure and the local price going into discount compared to the LME (after import duties and VAT are taken into consideration).

We expect consumption growth will remain positive and in excess (in tonnage terms) of local mine production growth. Moreover, our estimates see the 2009 inventory build in the few hundred thousands rather than millions of tonnes. As such, we expect that the need for material sourced out

of the country (whether this is in the form of concentrate or cathode) will remain strong for the foreseeable future, against a backdrop of uncertainty towards supply.

Physically backed ETF launches on the London Stock Exchange

On 10th December, the first ever physically backed copper ETF launched on the London Stock Exchange. The ETF Securities issued product, dubbed "ETFS Physical Copper", offers investors the ability to purchase approximately 5 kilograms of physical copper per share (the weight-per-share figure declines over time, to account for fees and charges). In line with GFMS' expectations, the fees charged for the product are considerably higher than those for the physical precious metals ETFs and comprise a 0.69% management fee a 0.12% insurance fee and a 36 cents per tonne, per day rental (storage) fee. At the time of writing, the data available at ETF Securities' web site suggested total holdings of the product stood at around 1,445 tonnes.

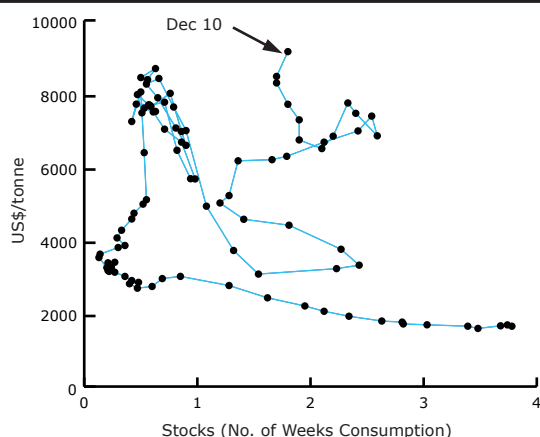
Collahuasi port closes after accident

An accident at the main port through which concentrate from Collahuasi is shipped marked the beginning of the new chapter in Xstrata's woes in Chile. The accident occurred less than two weeks following the resolution of a month-long strike and saw the shiploader collapse. The company declared force majeure, although it stated it continued production at a normal pace. Given the nature of the damage, it may be some time until the port can be used again. Xstrata reportedly started shipping material from Antofagasta's port, although it is not to-date clear if force majeure will be lifted.

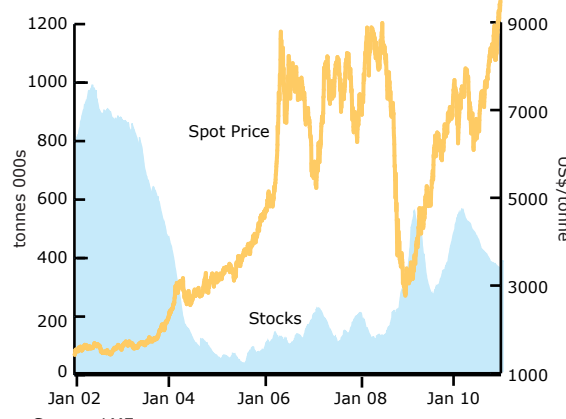
2010 Chilean production set to prove flat year-on-year

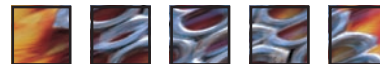
Another year-on-year decline in Chilean mine production was noted, although this was only marginal, to a monthly figure of around 467,000 tonnes. Overall, production during January-November totalled 4.9 million tonnes and was up by 0.1% on the 2009 figure. The figures are increasingly

LME Stocks vs Spot Price



Copper Daily Stocks vs Price





suggestive of output for the year overall falling short of GFMS' already cautious expectations, that had seen output rise by 0.5% year-on-year, the 32-day strike at Collahuasi being a key source of this shortfall.

Elsewhere, Peruvian production over the first ten months of last year fell by 2.5% year-on-year to just over one million tonnes. In China, CNIA data suggested concentrates production in January-November was up by a fifth year-on-year. GFMS will assess our estimates for 2010 over the next couple of months, as companies announce their production results, and will release our final figures in forthcoming editions of our One Year Forecast as well as in our annual

Copper Survey 2011, which will be published in early April. For more information on the report and to order a copy, please contact charles.demeester@gfms.co.uk.

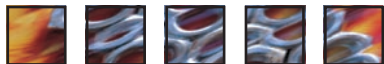
Market outlook

The general themes underpinning our projections for copper supply and demand over the next twelve months have not changed since the previous edition of our Base Metals One Year Forecast. We continue to be optimistic towards consumption growth for the year overall, primarily due to our expectation that, although slower than in previous years, demand growth in China will keep to healthy levels during the year and will be compounded by strength in

Copper Supply-Demand Balance 2005-2011

(000 tonnes)	2005	2006	2007	2008	2009	2010	2011
Global Production	16,572	17,291	17,934	18,200	18,367	18,929	19,602
% change y-o-y	4.0	4.3	3.7	1.5	0.9	3.1	3.6
Consumption							
Europe	3,887	4,255	4,096	3,758	3,271	3,638	3,684
Japan	1,229	1,282	1,252	1,185	876	975	990
United States	2,566	2,468	2,352	2,143	1,655	1,760	1,785
Total Mature	7,681	8,005	7,701	7,086	5,802	6,373	6,459
% change y-o-y	-6.7	4.2	-3.8	-8.0	-18.1	9.9	1.3
Brazil	335	339	332	372	319	355	363
China	3,690	3,688	4,885	5,147	6,421	7,000	7,490
India	373	401	511	507	573	640	685
Russia	635	692	671	650	377	390	395
Total BRICs	5,032	5,119	6,398	6,677	7,690	8,385	8,933
% change y-o-y	9.7	1.7	25.0	4.4	15.2	9.0	6.5
ASEAN	715	780	772	753	687	750	773
% change y-o-y	1.1	9.0	-1.0	-2.4	-8.8	9.1	3.1
South Korea	791	827	832	814	886	895	902
Taiwan	635	638	603	582	504	535	540
Others	1,747	1,770	1,847	1,981	1,901	2,059	2,128
Global Consumption	16,601	17,139	18,152	17,893	17,471	18,997	19,734
% change y-o-y	-1.2	3.2	5.9	-1.4	-2.4	8.7	3.9
Metal Balance	-29	152	-218	307	896	-68	-132
Reported stock change	-72	263	-103	134	273		
Reported Stocks							
Producer stocks	601	750	630	606	612		
Merchant stocks	6	6	15	25	15		
Consumer stocks	104	122	134	129	102		
Exchange Stocks							
Comex	7	34	15	35	99		
Shanghai Exchange	58	28	26	18	95		
LME	92	191	199	341	502		
Total Stocks	868	1,131	1,028	1,163	1,436	1,266	1,135
Total as No. weeks con	2.7	3.4	2.9	3.4	4.3	3.5	3.0
LME as No. weeks con	0.3	0.6	0.6	1.0	1.5		
LME cash (\$/tonne)	3,864	6,731	7,126	6,952	5,164	7,539	9,000
% change y-o-y	34.7	74.2	5.9	-2.4	-25.7	46.0	19.4

Source: GFMS, ICSG, WBMS, LME



other emerging markets. It is also worth noting that we are increasingly aware of conditions in mature markets proving better than had generally been expected.

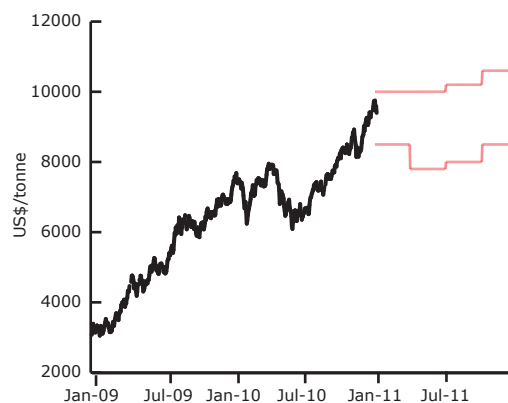
Moving to production, we continue to stand by our conservative projections. The recent developments in Chile provide further support to our view that the risks to the downside exceed those to the upside, despite the extremely attractive price environment for the industry.

As a result, we continue to forecast a market in deficit for 2011, the magnitude of which will be considerably higher than the one seen last year. Although our expectations for the market's fundamentals are broadly unchanged from one month ago, we have made noteworthy changes to our price projections.

We continue to see prices suffering a significant correction from current levels at some point within the first half of the year, but given the higher starting point and the precedent of prices close to \$10,000/tonne having been set, we no longer believe that a drop to levels in the low-\$7,000s are realistic. The lower end of our projected trading range over the next six months has been set to a little less than \$8,000/tonne.

Moreover, we now see the recovery in the second half of the year to see prices far exceed the levels we had previously forecast under our **Base Case** and indeed match what had previously represented our more bullish **Scenario B**. Specifically, we see prices peaking in the mid-\$10,000s before the end of the year and averaging above \$9,000/tonne over the second half, resulting in a full-year average of \$9,000/tonne.

Copper Price & Forecast Trading Range



Source: Thomson Reuters EcoWin, GFMS

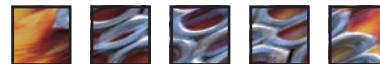
Naturally, we have pushed upwards our price-projections under our high case **Scenario B** forecast somewhat. It is worth noting that our projections under this scenario are only marginally higher than those of our Base Case, a reflection of the latter being already quite bullish and also of our view that any further upside is limited, even under our most bullish set of assumptions. Although given the volatile nature of the markets we cannot rule out ventures towards the \$12,000/tonne barrier, for instance, we would expect this to result in heavy profit-taking and demand destruction which, coupled with a more limited reaction by production, would culminate in a prompt correction.

In contrast, were our most bearish (and in our view least likely) **Scenario C** to materialise, prices could post very dramatic declines, given the gap between current price levels and production costs. As such, our projections under Scenario C are only slightly higher than those featured in last month's edition, reflecting the higher starting point.

Quarterly Copper Supply-Demand Balance - Base Case

	2010				2011			
(000 tonnes)	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Global Production	4,610	4,700	4,760	4,860	4,730	4,890	4,955	5,027
% change y-o-y	5.4	4.5	1.6	1.1	2.6	4.0	4.1	3.4
Global Consumption	4,537	4,650	4,800	5,010	4,700	4,870	5,050	5,114
% change y-o-y	13.1	6.9	6.2	9.2	3.6	4.7	5.2	2.1
Market Balance	73	50	-41	-151	30	20	-95	-87
Reported Stocks	1,481	1,390	1,238	1,266	1,296	1,316	1,221	1,134
No. of weeks' consumption	4.2	3.9	3.4	3.3	3.6	3.5	3.1	2.9
LME Cash \$/tonne	7,243	7,013	7,242	8,634	9,000	8,400	9,100	9,500
% change y-o-y	110.9	50.0	24.0	30.0	24.3	19.8	25.7	1.0
Range								
High	7,830	7,951	8,054	9,740	10,000	10,000	10,200	10,600
Low	6,242	6,091	6,354	8,086	8,500	7,800	8,000	8,500

Source: GFMS, WBMS, ICSG, LME

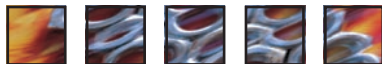


Statistical Appendix

Copper Mine Production 2005-2011

(000 tonnes)	2005	2006	2007	2008	2009	2010	2011
Zambia	433	474	507	539	615	750	825
Other Africa	238	265	326	430	494	535	562
Total Africa	671	739	833	969	1,110	1,285	1,386
China	777	889	946	1,022	1,041	1,160	1,237
Indonesia	1,065	816	789	653	972	821	685
Kazakhstan	402	434	456	459	468	444	437
Other Asia	513	594	622	636	709	741	783
Total Asia	2,756	2,733	2,813	2,771	3,189	3,166	3,141
Poland	512	497	452	429	441	425	425
Russia	510	510	511	512	512	513	514
Other Europe	536	566	586	608	623	678	703
Total Europe	1,558	1,573	1,549	1,549	1,576	1,616	1,642
Canada	596	603	588	606	489	496	532
Chile	5,321	5,361	5,555	5,299	5,388	5,416	5,678
Mexico	429	334	320	244	242	329	435
Peru	1,010	1,049	1,156	1,228	1,224	1,212	1,181
United States	1,157	1,222	1,178	1,330	1,206	1,179	1,161
Other Americas	320	324	381	374	370	383	394
Total Americas	8,832	8,893	9,178	9,081	8,919	9,014	9,379
Australia	916	859	873	880	838	811	819
New Caledonia	193	194	169	160	167	183	186
Total Oceania	1,109	1,053	1,042	1,039	1,004	994	1,005
Global Total	14,925	14,991	15,415	15,410	15,799	16,075	16,554
Western World	12,237	12,122	12,510	12,432	12,785	12,894	13,281
Former Socialist	2,687	2,869	2,905	2,978	3,014	3,181	3,272

Source: GFMS, ICSG



Refined Copper Production 2005-2011

(000 tonnes)	2005	2006	2007	2008	2009	2010	2011
South Africa	99	100	111	93	89	91	95
Zambia	399	418	431	417	414	450	510
Other Africa	31	35	47	72	169	231	301
Total Africa	529	553	589	582	672	772	906
China	2,600	3,003	3,499	3,795	4,121	4,400	4,650
India	493	625	715	662	715	725	735
Indonesia	263	218	277	254	289	295	305
Japan	1,395	1,532	1,577	1,540	1,440	1,470	1,490
Kazakhstan	418	428	406	398	368	375	405
Philippines	172	181	160	175	178	182	185
South Korea	519	576	582	531	539	542	545
Other Asia	510	538	510	455	433	516	540
Total Asia	6,370	7,100	7,726	7,809	8,083	8,505	8,855
Belgium	386	391	394	394	373	370	380
Germany	638	662	666	690	669	670	675
Poland	560	557	533	527	503	515	520
Russia	935	951	949	862	874	865	860
Scandinavia	394	407	358	396	345	350	355
Spain	309	299	308	319	335	360	390
Other Europe	236	258	252	329	374	391	400
Total Europe	3,458	3,525	3,460	3,517	3,472	3,521	3,580
Brazil	199	220	219	227	213	230	245
Canada	515	501	454	442	336	350	360
Chile	2,824	2,811	2,936	3,058	3,272	3,200	3,250
Mexico	415	369	357	295	261	280	280
Peru	510	508	414	463	423	420	420
United States	1,255	1,250	1,311	1,280	1,161	1,200	1,210
Other Americas	26	26	26	26	27	26	26
Total Americas	5,745	5,684	5,717	5,791	5,693	5,706	5,791
Total Oceania	469	429	442	502	446	425	470
Global Total	16,572	17,291	17,934	18,200	18,367	18,929	19,602
Western World	11,825	12,135	12,332	12,368	12,166	12,425	12,812
Former Socialist	4,747	5,156	5,603	5,832	6,201	6,504	6,790

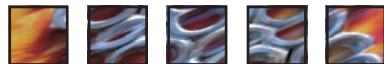
Source: GFMS, ICSG



Refined Copper Consumption 2005-2011

(000 tonnes)	2005	2006	2007	2008	2009	2010	2011
Total Africa	228	235	270	297	304	323	332
China	3,690	3,688	4,885	5,147	6,421	7,000	7,490
India	373	401	511	507	573	640	685
Japan	1,229	1,282	1,252	1,185	876	975	990
Malaysia	180	189	184	177	167	178	182
South Korea	791	827	832	814	886	895	902
Taiwan	635	638	603	582	504	535	540
Thailand	241	254	250	240	215	232	235
Turkey	327	321	388	376	325	348	358
Other Asia	714	839	848	864	886	986	1,028
Asia Total	8,180	8,438	9,751	9,892	10,854	11,789	12,410
Belgium	269	291	309	285	234	272	274
France	505	540	440	410	310	290	295
Germany	1,179	1,416	1,408	1,391	1,133	1,320	1,330
Italy	667	790	743	618	563	620	630
Poland	275	275	297	247	221	237	240
Russia	635	692	671	650	377	390	395
Scandinavia	255	266	259	237	209	233	238
Spain	321	319	330	385	338	350	353
United Kingdom	200	180	55	54	49	51	52
Other Europe	196	177	241	231	309	363	371
Total Europe	4,502	4,945	4,752	4,507	3,743	4,126	4,178
Brazil	335	339	332	372	319	355	363
Canada	290	301	206	197	142	155	160
Mexico	430	360	340	325	277	305	318
United States	2,276	2,168	2,146	1,946	1,513	1,605	1,625
Other Americas	202	208	205	203	186	197	202
Total Americas	3,533	3,376	3,229	3,044	2,438	2,617	2,668
Total Oceania	155	143	148	151	130	140	145
Global Total	16,601	17,139	18,152	17,893	17,471	18,997	19,734
Western World	11,778	12,171	11,924	11,468	10,085	10,965	11,192
Former Socialist	4,823	4,967	6,228	6,424	7,386	8,032	8,542

Source: GFMS, ICSG



Nickel

Recent developments

- **Nickel staged an impressive performance over December, with the monthly average rising by 5.2% month-on-month to \$24,111/tonne. A good end to the year was followed by further strengthening as the price of LME spot nickel rose to a 7-month high of \$25,175/tonne on the first trading day of the year, noticeably in line with the rest of the base metals complex.**
- **Meanwhile, LME stocks have continued to increase although recent days have seen the rise in inventories decelerate. At just above 137,000 tonnes in early January, they are around 4.5% higher than a month ago.**
- **On the supply-side, Anglo American's 36,000 tonne per year Barro Alto ferro-nickel project is due to start up this quarter. The company expects the plant to ramp up to full production in the second half of 2012.**
- **According to the CNI-A, Chinese refined nickel production fell for the second month in a row to 13,803 tonnes in November. Within this, Jinchuan produced 10,100 tonnes.**
- **Chinese imports of refined nickel dropped by a third in November to 12,044 tonnes, notably the second lowest level seen in 2010.**
- **On the demand-side, the WBMS estimates the global nickel consumption increased by almost 13.0% year-on-year to 1.24 million tonnes up to October, despite demand having remained down by over 10% since July.**

Nickel Supply-Demand & Price Forecast

	2011			
(000 tonnes)	Q1	Q2	Q3	Q4
Supply/Demand Forecast (Base Case)				
Production	380	385	391	410
Consumption	377	390	383	397
Balance	3	-5	8	13
Stocks	142	137	145	158
Price Forecast (\$/tonne)				
Base Case	24,495	24,997	23,986	22,480
Scenario B	26,000	26,500	25,500	26,000
Scenario C	22,000	21,000	20,500	19,500

Source: GFMS

Outlook for the next 12 months

- **In line with market developments, GFMS have revised their supply-demand balance projections for both 2010 and 2011. Given the stock build on the LME over the fourth quarter, we have revised downwards our deficit projection for 2010 to 37,000 tonnes.**
- **For 2011, we are now projecting the market in a surplus of 19,000 tonnes. With the bulk of the demand side recovery behind us, this will be driven by the supply-side. We expect higher output from Vale's Canadian operations and strong growth in Chinese NPI production. We also expect ferro-nickel projects, namely Barro Alto, Onca Puma and Tagaung Taung, to see the supply-side higher this year.**
- **Taking into consideration nickel's recent price performance and our belief that nickel will continue rising strongly along with the rest of the complex, GFMS are forecasting prices to average \$24,495/tonne in the first quarter before rising further in the second quarter on traditionally strong demand. However, this will be followed by declines as supply gains momentum. As a result, we forecast an average annual price of \$23,990/tonne this year.**

GFMS' Forecast Scenarios

Base Case, 50% Probability

Represents what GFMS consider the most likely outcome for the markets.

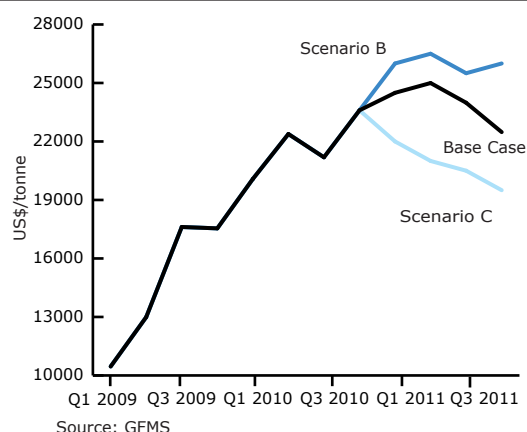
Scenario B, 30% Probability

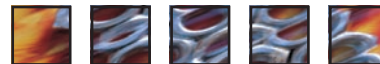
Faster recovery than under our Base Case in the near-term and stronger growth thereafter.

Scenario C, 20% Probability

The recovery that has been seen in the mature economies thus far slows markedly while growth in China continues to moderate as the impact of the stimulus packages wear off.

Quarterly Nickel Price & Forecast





Market background

Nickel prices have started the New Year on a high note, rising to a level not seen since May 2010. The cash quote which breached \$25,000/tonne on the first trading day of the year was also a notable 34% higher than a year ago. Indeed, this follows the metal's impressive performance over December, which saw the average cash quote rise by 5.2% to \$24,111/tonne. This rally, which has been seen across the base metals complex, has largely been led by copper, which in turn has been driven by strong investor interest.

Meanwhile, LME stocks which highlight the metals weak fundamentals have continued to trend upwards. Although they finished the year 13.6% lower year-on-year, they rose by around 4.5% since the last time we reported them. They are currently just over 137,000 tonnes in early January.

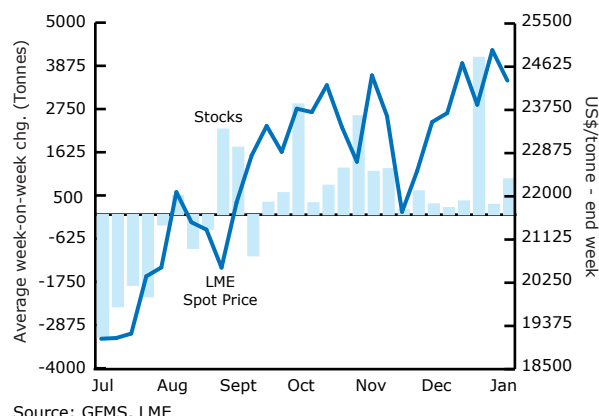
The supply-side continues to advance...

There have been a handful of supply-side developments since our last Base Metals One Year Forecast, which leads us to continue forecasting exceptional refined production growth this year. On the mine-side, there have been reports that Norilsk Nickel will be restarting production at its Lake Jonhston mine in Australia in the first half of the year. The concentrate produced will be sent to the company's Harjavalta plant in Finland for refining. The company has also been granted two exploration licenses by the government of Botswana as the nickel miner ramps up its activities to extend the life of its Tati nickel mine to beyond 2016.

Similarly, construction at the Kennecott Eagle mine in the United States remains on target after the facility received approval to extend its electrical power infrastructure. The mine is scheduled to begin in late 2013 with nickel output estimated at 16,000 tonnes.

Also, Mirabela Nickel is set to achieve and possibly even exceed its 2010 production guidance of 9,500-10,000 tonnes of nickel contained in concentrate as higher output was recorded at the company's Santa Rita mine in Brazil. The

Weekly Change in LME Stocks vs Spot Price



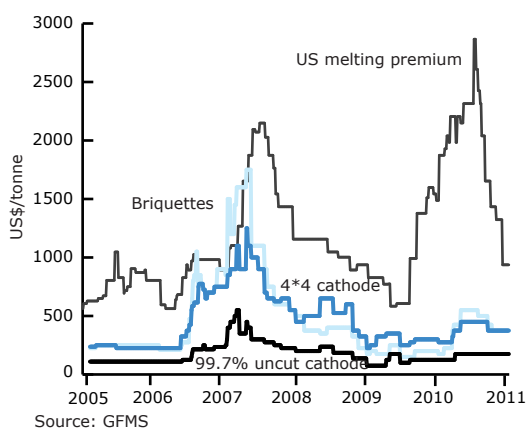
mine which commenced in late 2009 is expected to reach full capacity of 23,000 to 25,000 tonnes in 2012. Completion of the open pit pre-strip and the expansion of the processing plant are planned for 2011. The whole of Santa Rita's concentrate production is sold until the end of 2014 to both Votorantim and Norilsk Nickel.

In terms of refined production, Anglo American's 36,000 tonne per year Barro Alto ferro-nickel project also in Brazil is on track to start up in the first quarter of 2011 as originally planned. The company expects the plant to ramp up to full production in the second half of 2012. In the first five years, output is expected to average 41,000 tonnes per year of nickel contained in ferro-nickel, while over the 26 years of the life of the plant, production will average 36,000 tonnes per year. Anglo is also studying two potentially world class projects in Brazil, Jacare and Morro Sem Bone, which could add some 66,000 tonnes per year of nickel in the future. A possible start-up for the latter two is 2015.

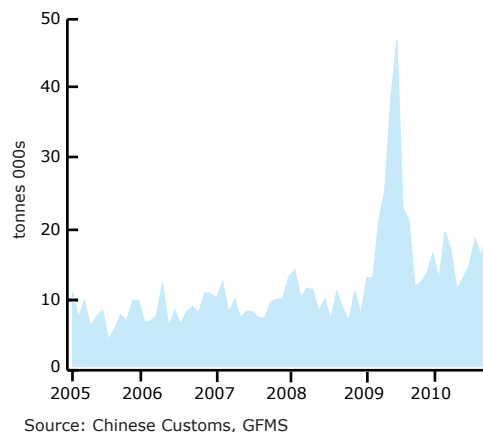
...as China continues to underpin global refined production

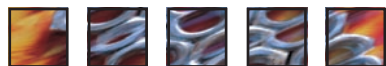
The latest production data by the World Bureau of Metal Statistics (WBMS), which only stretches to October, continues to point to rising nickel supply. According to the

Nickel Premiums



Chinese Nickel Imports





organisation, global refined output increased by 12.4% year-on-year to just short of 1.23 million tonnes over the first ten months of the year, and follows almost double-digit growth over January-September. A regional breakdown however shows that the main contributor to this growth was Asia, which saw its output increase by almost 40% year-on-year unsurprisingly on the back of China (+53%).

Strong growth in the country has largely been due to rising nickel pig iron (NPI) production, which has not only been encouraged by higher nickel prices but also surging Chinese imports of nickel ore. The latter, which reflects strong demand from NPI producers, surged to new all-time high in November, which took the cumulative total to 22.46 million tonnes, some 50% higher than in 2009. Of the 2.83 million tonnes shipped in the month, Indonesia supplied slightly less than half the ore, while Philippines shipped around 53%.

On the other hand, official data by the China Nonferrous Metals Industry Association (CNI-A) suggests that Chinese refined nickel production (which excludes output from NPI producers) slipped for the second consecutive month to 13,803 tonnes in November. This placed output over the first eleven months 2.8% lower than in the same period in 2009 at 150,858 tonnes. Within the November total, Jinchuan produced 10,100 tonnes bringing its year-to-date total to 117,800 tonnes, just over 12,000 tonnes short of its annual target of 130,000 tonnes.

Global consumption maintains double-digit growth

Turning to the demand side of the market, the WBMS estimates that global nickel consumption increased by almost 13% year-on-year to 1.24 million tonnes up to October. This is despite demand having remained over 10% down from July's monthly level of 140,700 tonnes. Nevertheless, strong year-to-date performances from Western countries, particularly the United States (+50%) and Europe (+36.8%), has seen global consumption grow at healthy rates. However, it should be noted that the increases not only reflect robust growth over the first half of the year but

also a low 2009 base. Japanese demand growth also came in strong but was lower than the two above-mentioned countries at 27.8% year-on-year.

Chinese demand remains in negative territory

By comparison, Chinese demand fell by 4.2% over the first ten months. This is not because demand has weakened, but because of an extremely high base set by the record high imports over the first seven months of last year, when stocks were built up. Inventories have been drawn down so far this year.

The latest data by the CNI-A suggests that Chinese imports of refined nickel dropped by more than a third in November. Shipments to the country declined by 36.8% month-on-month to 12,044 tonnes, notably the second lowest level seen this year, following the 10,902 tonnes recorded in May. Nevertheless, imports over the eleven months totalled 167,015 tonnes, almost 30% down on last year.

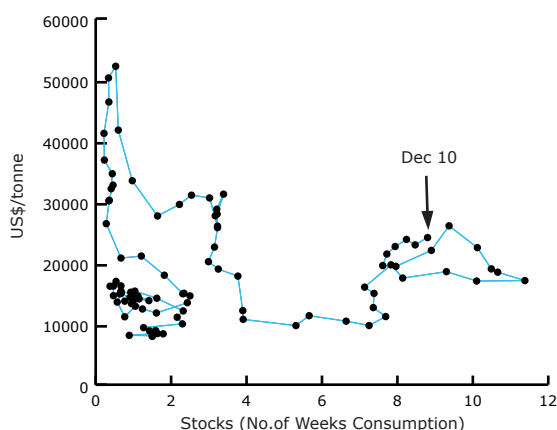
Contract premiums set to decline

Physical premiums for 2011 contracts are expected to fall to between 30-50 cents/lb in the United States, which is in sharp contrast to 2010 premiums of 70-80 cents/lb. Estimates for next year's contract premiums come as Vale has been offering spot premiums for melting-grade briquetted as low as 24.0 cents/lb to win back customers after the year-long strike at its Sudbury operations. Notably, contract premiums are following those for spot melting-grade material, which have slid by almost 100% to between 35-50 cents/lb from \$1.20-1.40/lb in July.

Poor stainless demand in Europe takes its toll

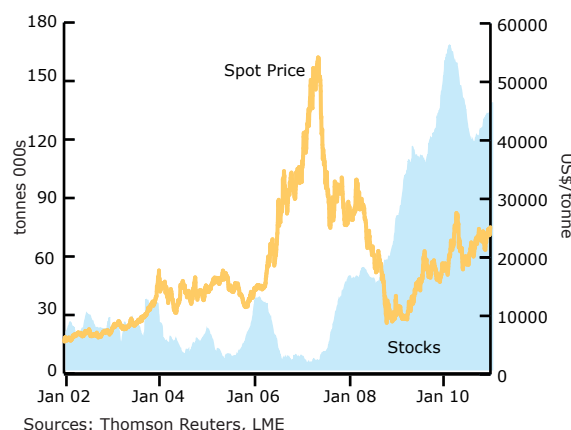
Arcelor Mittal's recent decision to idle its cold rolling stainless steel operation at Isbergues in northern France highlights Europe's weak demand conditions. The plant will restart only when the company is confident of a sustainable pick up in stainless demand. At the same time, the company will invest \$62 million on its annealing and pickling line at the Gueugnon site, which produces non-nickel ferritic stainless.

LME Stocks vs Spot Price

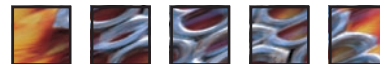


Source: GFMS, LME

Nickel Daily Stocks vs Price



Sources: Thomson Reuters, LME



Having said this, stainless steel coil producers, Outokumpu and Acerinox, have both announced base price increases for the first quarter of 2011 following a slight recovery in orders, helped by the likelihood of higher alloy surcharges at the start of the New Year.

Chinese stainless output set to pick up

The tightness in the Chinese stainless market that we had been reporting in recent months is set to ease as major producers prepare to return to normal production with the end of emissions controls and maintenance works. Taigang Steel is set to increase output by at least 30% this month compared to November. Previously, maintenance works over October and November cut production by 50,000 tonnes in total. Guangzhou Lianzhong is also preparing to return to normal production of 130,000 tonnes per month after partially shutting down for the Asian Games in November. The company is believed to have lost 30,000 tonnes of output as a result. Similarly, Baosteel Stainless expects its

production to pickup by 10-20% to around 120,000 tonnes in January, after wrapping up maintenance in December.

Market balance and price outlook

Taking into consideration the metal's relatively poor fundamentals and importantly nickel's recent price performance, GFMS have revised both their supply-demand balance projections and price forecasts.

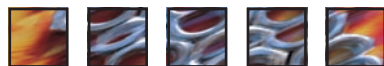
Previously, we were forecasting the market in a small deficit of 5,000 tonnes this year, which of course followed a somewhat larger deficit in 2010. However, in line with weak stainless steel production in the fourth quarter and the restart of Vale's Sudbury operations, which in turn saw LME stocks build up, GFMS have revised down last year's deficit from 58,000 tonnes to 37,000 tonnes.

With regards to 2011, we are now projecting the market to register a surplus of 19,000 tonnes. This is based mainly

Nickel Supply-Demand Balance 2005-2011

(000 tonnes)	2005	2006	2007	2008	2009	2010	2011
Global Production	1,288	1,336	1,388	1,353	1,327	1,429	1,566
% change y-o-y	2.9	3.7	3.9	-2.5	-2.0	7.7	9.6
Consumption							
Europe	441	458	428	397	237	311	333
Japan	180	181	196	185	148	176	182
United States	135	145	119	117	85	121	126
Total Mature	756	784	743	699	470	608	641
% change y-o-y	0.6	3.7	-5.2	-5.9	-32.8	29.5	5.3
Brazil	26	25	22	24	14	12	18
China	195	234	328	305	541	548	574
India	16	18	19	21	24	25	27
Russia	26	26	26	26	26	30	33
Total BRICs	263	303	395	377	606	615	652
% change y-o-y	25.0	15.3	30.3	-4.7	60.8	1.5	5.9
South Korea	118	93	71	76	93	100	104
Taiwan	84	107	76	69	64	74	78
Other	75	79	71	74	66	68	73
Global Consumption	1,296	1,366	1,356	1,294	1,299	1,466	1,547
% change y-o-y	3.6	5.4	-0.7	-4.5	0.4	12.8	5.6
Metal Balance	-8	-30	32	59	28	-37	19
Reported stock change	14	-25	38	30	75		
Reported stocks							
Country stocks	76	80	77	76	72		
LME	36	7	48	79	158		
Total Stocks	112	87	125	155	230	193	212
Total as no. weeks' consumption	4.5	3.3	4.8	6.2	9.2	6.9	7.1
LME as no. weeks' consumption	1.4	0.3	1.8	3.2	6.3		
LME Cash (\$/tonne)	14,733	24,287	37,181	21,029	14,700	21,812	23,990
% change y-o-y	6.4	64.9	53.1	-43.4	-30.1	48.4	10.0

Source: GFMS, WBMS, LME

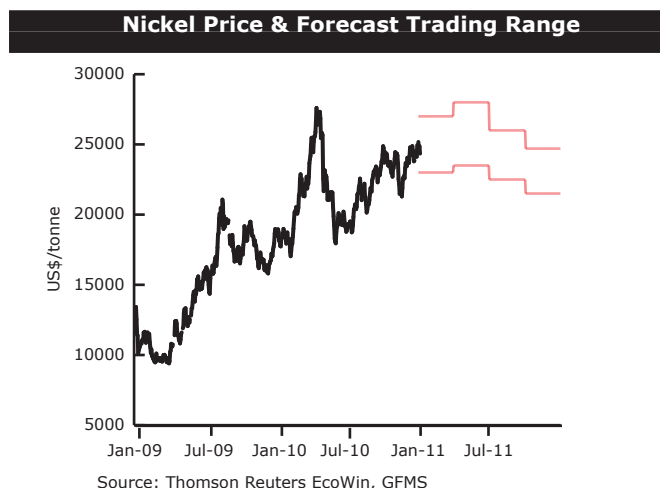


on higher output from Vale's Candian operations (assuming a restart at Voisey's Bay), the restart of Falcondo, as well as new supply that is expected to come online. While we are uncertain as to whether the new batch of HPAL projects will start-up on time, which is why we have kept their contributions minimal, we believe that additional output from ferro-nickel projects, Barro Alto, Onca Puma and Tisco's Tagaung Taung, will see the supply-side higher this year. We also expect strong growth in Chinese nickel pig iron production, as prices remain above the marginal cost of production for these producers. Therefore, GFMS are forecasting global refined output to increase by 9.6% this year. On the demand-side, we expect global consumption to increase by 5.6% in 2011, following the sharp rebound last year.

With prices having finished 2010 on an extremely positive note, noticeably 32% higher than at the start of the year, and the fact that nickel has been rising strongly recently, GFMS have taken a bullish stance towards pricing. We have revised our first quarter price forecast upwards to \$24,495/tonne, which is largely based on the strength across the board that we foresee rather than a reflection of the metal's own fundamentals. In the second quarter, we expect prices to pick up further on reduced output at Cerro Matoso and traditionally strong nickel demand, which in turn should see the average increase to just short of \$25,000/tonne. However, as we move into the final two quarters of the year, increasing supply (as mentioned above) should see prices decline, but will nevertheless remain above the 2010 annual average of \$21,812/tonne. Therefore, we are forecasting an annual average price of \$23,990/tonne this year.

Alternative scenarios

For prices to fall to our low case forecast (**scenario C**) of \$20,750/tonne, a notable 13.5% below our base case forecast, we expect the much anticipated HPAL projects to

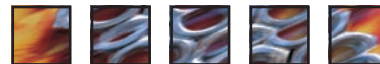


come online as scheduled. Given that this technology has been tried-and-tested for so long now (in sharp contrast to the first generation of HPAL projects), there is little reason for these projects not to materialise. We also expect economic conditions particularly in Europe to worsen as the sovereign debt crisis continues to weigh on the region.

Scenario B, however is predicated on simply a reversal of the above. We expect a noticeable and sustained recovery in the global economy, which in turn should see global nickel demand improve faster than in our base case. On the supply-side however, other than our belief that the start-up dates of HPAL projects will be pushed back, we expect limited output from Chinese NPI producers. The main impediment to this will be continued high grade nickel ore from Indonesia. With the government having recently imposed new restrictions on the export of ores containing more than 35% moisture and reports of potential export taxes for Indonesian ore exports, this in turn could significantly raise operating costs for these producers. Under these circumstances, we expect prices to reach our high case forecast of \$26,000/tonne.

Quarterly Nickel Supply-Demand Balance - Base Case								
	2010				2011			
(000 tonnes)	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Global Production	337	361	354	377	380	385	391	410
% change y-o-y	8.7	9.6	4.1	8.5	12.8	6.6	10.5	8.8
Global Consumption	343	386	371	366	377	390	383	397
% change y-o-y	19.4	19.5	9.0	3.0	10.0	1.0	3.2	8.5
Market Balance	-6	-25	-17	11	3	-5	8	13
Reported Stocks	170	145	128	139	142	137	145	158
No. of weeks' consumption	6.5	4.9	4.5	5.0	4.9	4.6	4.9	5.2
LME Cash \$/tonne	20,078	22,382	21,188	23,598	24,495	24,997	23,986	22,480
% change y-o-y	92.0	72.3	20.3	34.5	22.0	11.7	13.2	-4.7
Range								
High	24,950	27,600	23,425	24,960	27,000	28,000	26,000	24,700
Low	17,035	17,955	18,375	21,290	23,000	23,500	22,500	21,500

Source: GFMS, WBMS, LME



Cobalt

Market background

We have noted in previous reports that the second half rally seen in the LME metals largely by-passed the cobalt market. Values have posted only limited gains in the final quarter. However prices did finish the year on a firmer note at \$17.25/lb. High grade material ended December at \$18.95 compared to \$17.0/lb at the beginning of the December. Further advances were seen in early 2011.

The latest trade data implies that purchasing in China was quiet in so far in the final quarter. Imports of cobalt concentrate in October fell by nearly 50% to a fourteen-month low of 17,411 tonnes. There was a pick up in November to 26,393 tonnes and reflecting the high level of imports earlier this year, which has led to the build up of inventory, cumulative imports of cobalt concentrate rose by 61% year-on-year up to November to 317,457 tonnes.

The Jinchuan Group recently raised its cobalt cathode price by 3.2%, while chloride producers also increased prices. Although we are entering a period of seasonally strong demand prior to the Chinese New Year, it is believed that inventories within the country are relatively high. This is likely to limit purchases. It is reported that some of the major cobalt-containing battery material producers purchased significant amounts of material in late 2010 when prices were low and therefore have built up inventory.

The battery market should support demand (long term)

Although current demand conditions are poor, a number of announcements have been made concerning future growth prospects in the automotive sector. According to the National Battery Power Testing Center in China, the country's hybrid electric vehicle output may reach 500,000

Cobalt Supply-Demand & Price Forecast				
	2011			
(000 tonnes)	Q1	Q2	Q3	Q4
Supply/Demand				
Forecast (Base Case)				
Supply	18,200	18,750	19,500	21,060
Consumption	16,350	17,250	16,250	17,511
Balance	1,850	1,500	3,250	3,549
Price Forecast (\$/tonne)				
Base Case	18.2	16.9	16.0	15.0
Scenario B	21.0	22.0	22.0	24.0
Scenario C	16.0	15.0	14.0	14.0

units by 2012, which would require an estimated 2.5bn KWh of battery power. China's Ministry of Science and Technology have ambitious projections suggesting that electric vehicle output might reach 1m units by 2020.

Supply growth keeps a cap on prices

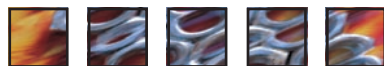
Despite the underlying disputes concerning mining rights within the DRC, cobalt production within the country continues to increase. According to Freeport McMoRan, production at Tenke, which began in March 2009 should reach 18m lbs by 2011. Zambian cobalt production increased to 5,579 tonnes in the first eight months of 2010, up from 3,180 tonnes in the corresponding period of last year.

The Cobalt Development Institute has released its estimates of cobalt metal production for the first six months of 2010. It shows strong growth from its members and from other suppliers. Output at its members increased by 18% year-on-year to 14,317 tonnes. A significant part of the increase can be attributed to the restart at Chambishi as the company produced 1,579 tonnes compared to 34 tonnes a year earlier. Other major contributors where QNPL in Australia, where

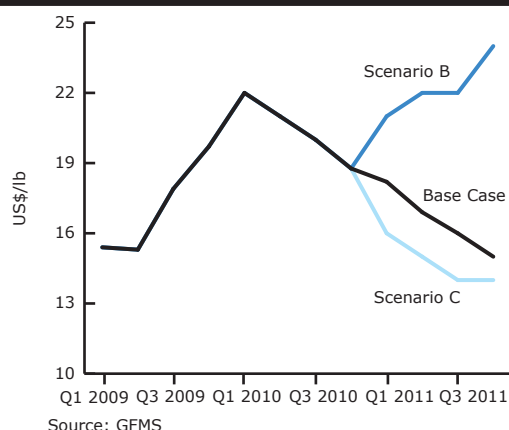
Cobalt Supply-Demand Balance 2005-2011

(tonnes)	2005	2006	2007	2008	2009	2010	2011
Global Production	50,647	52,064	53,013	57,040	52,940	65,770	77,510
DLA Stockpile sales	1,199	294	617	100	150	5	0
Global Supply	51,846	52,358	53,630	57,140	53,090	65,775	77,510
% change y-o-y	4.1	1.0	2.4	6.5	-7.1	23.9	17.8
Global Consumption	50,892	52,602	54,817	55,454	54,921	61,983	67,361
% change y-o-y	4.8	3.4	4.2	1.2	-1.0	12.9	8.7
Market balance	954	-244	-1,187	1,686	-1,831	3,792	10,149
Cobalt 99.8% \$/lb	16.1	16.7	29.2	39.2	17.3	21.0	16.5
% change y-o-y	-33.8	4.1	75.0	34.1	-55.9	21.4	-21.4
Cobalt 99.3% \$/lb	14.8	15.7	28.0	36.7	15.8	19.1	15.5
% change y-o-y	-36.1	6.1	78.3	31.1	-56.9	20.9	-18.8

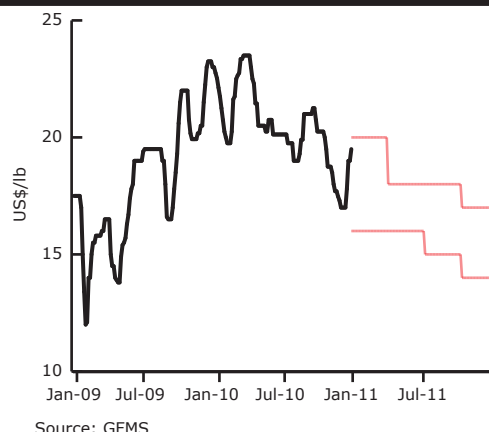
Source: GFMS, CDI, WBMS, Metal-Pages



Quarterly Cobalt Price & Forecast (99.8%)



Cobalt Price & Forecast Trading Range (99.8%)



output was 1,043 tonnes, up from 600 tonnes a year earlier and Sumitomo, which has been treating the additional feed from Coral Bay, with production near tripling from 1,401 tonnes from 507 tonnes.

There was a much larger increase from non-CDI members with output up to an estimated 22,129 tonnes from 12,319 tonnes a year earlier. Although China dominated with an increase from 15,532 tonnes, up from 6,926 tonnes. Significant increases were registered at the Katanga operation in the DRC and Votorantim in Brazil. The bottom line is an increase in production of close to 50% in the first six months of the year. Although we believe that the Chinese figure may overstate the increase, it is clear that the market has had to absorb a lot of extra metal.

...with further increases in the pipeline

The cobalt market will also have to absorb higher output associated with the growth in refined nickel production reflecting higher utilization rates at existing operations and new capacity. Production should rise sharply at Vale following the resolution of the strike at Sudbury. It appears that the strike at the company's Voisey's Bay operation is finally reaching its conclusion. Vale is also in the process of

ramping up its Goro project that could eventually yield 4,000 tpy of cobalt. Other contributors to extra cobalt production in 2011 should include Sherritt Gordon's 5,600 tpy Ambatovy project. Despite numerous delays the company still believes that the first metal will be produced in the summer. The market will also have to absorb projects from the African cobalt-belt the largest of which is Tenke.

Market and price outlook

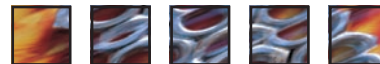
Although we remain positive about the demand prospects for cobalt, the market is having to cope with a surge in supply. The increase to supply is coming from a variety of sources – the reactivation of capacity hit by strikes, the expansion to existing capacity and, most notably, a large number of greenfield projects.

The new capacity is coming from a large number of projects and they are being developed by mainstream nickel-cobalt producers. As such, we are not expecting a repeat of the problems associated with the first generation of high pressure acid leach operations, when actual production was massively below expectations. Therefore we continue to forecast that the market will be in a surplus in 2011. (See supply demand balance tables for details).

Quarterly Cobalt Supply-Demand Balance - Base Case

	2010				2011			
(000 tonnes)	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Global Production	15,350	16,000	16,325	18,100	18,200	18,750	19,500	21,060
DLA Stockpile Sales	5	0	0	0	0	0	0	0
Global Supply	15,355	16,000	16,325	18,100	18,200	18,750	19,500	21,060
% change y-o-y	16.0	21.7	25.1	32.6	18.5	17.2	19.4	16.4
Global Consumption	15,250	16,250	14,500	15,983	16,350	17,250	16,250	17,511
% change y-o-y	17.8	23.1	9.4	16.1	7.2	6.2	12.1	9.6
Market Balance	105	-250	1825	2117	1850	1500	3250	3549
Cobalt 99.8% \$/lb	21.8	21.4	20.0	18.8	18.2	16.9	16.0	15.0
% change y-o-y	41.5	39.5	11.7	-4.8	-16.5	-20.8	-20.0	-20.1
Cobalt 99.3% \$/lb	20.1	19.5	18.0	17.4	17.2	15.9	15.0	14.0
% change y-o-y	51.2	35.1	9.1	-5.1	-14.5	-18.3	-16.7	-19.6

Source: GFMS, CDI, WBMS

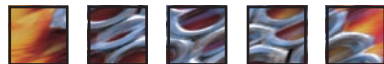


Statistical appendix

Refined Nickel Production 2005-2011 (000 tonnes)

(000 tonnes)	2005	2006	2007	2008	2009	2010	2011
South Africa	42	42	38	32	34	35	35
Total Africa	58	55	52	43	44	44	55
China	95	132	160	171	247	348	388
Japan	164	152	161	156	144	158	164
Other Asia	7	15	19	20	36	41	43
Total Asia	267	299	340	347	426	547	595
Finland	39	47	55	51	41	43	45
Norway	85	82	88	89	89	92	94
CIS	281	290	288	285	262	270	274
United Kingdom	38	37	34	39	18	26	30
Other Europe	40	42	51	51	40	48	51
Total Europe	482	498	515	514	450	479	494
Brazil	30	31	32	29	27	30	60
Canada	140	147	154	168	117	96	130
Colombia	53	51	49	42	52	50	38
Cuba	44	44	43	35	31	32	34
Dominican Republic	29	30	29	19	0	0	9
Total Americas	312	319	322	303	238	220	285
Australia	122	116	114	108	131	115	120
New Caledonia	47	49	45	37	38	39	52
Oceania	169	165	159	145	169	154	172
Global Total	1,288	1,336	1,388	1,353	1,327	1,429	1,566
Western World	869	870	897	863	786	779	870
Former Socialist	420	465	490	491	540	650	696

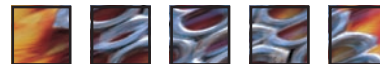
Source: GFMS, WBMS



Refined Nickel Consumption 2005-2011

(000 tonnes)	2005	2006	2007	2008	2009	2010	2011
Total Africa	48	55	45	45	44	43	46
China	195	234	328	305	541	548	574
India	16	18	19	21	24	25	27
Japan	180	181	196	185	148	176	182
South Korea	118	93	71	76	93	100	104
Taiwan	84	107	76	69	64	74	78
Other Asia	17	14	13	17	15	16	17
Total Asia	609	647	703	673	886	939	982
Belgium	49	58	55	47	15	20	22
Finland	49	51	40	41	17	33	35
Germany	116	106	110	91	62	92	99
Italy	60	68	64	68	44	52	56
Spain	48	53	42	41	24	28	31
Other Europe	145	149	143	135	101	116	122
Total Europe	468	484	455	423	264	341	365
United States	135	145	119	117	85	121	126
Other Americas	35	33	32	34	19	19	26
Total Americas	170	178	151	151	104	140	152
Global Total	1,296	1,366	1,356	1,294	1,299	1,466	1,547
Western World	1,055	1,085	980	941	713	867	921
Former Socialist	240	281	376	353	586	598	627

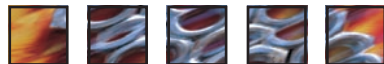
Source: GFMS, WBMS



Cobalt Supply 2005-2011

(tonnes)	2005	2006	2007	2008	2009	2010	2011
Morocco	1,613	1,405	1,591	1,600	1,610	1,610	1,600
South Africa	214	257	275	280	280	800	2,500
Uganda	600	600	600	600	650	660	660
DRC	600	550	606	700	2,000	4,000	4,000
Zambia	5,422	4,665	4,435	4,600	3,000	5,000	5,750
Total Africa	8,449	7,477	7,507	7,780	7,540	12,070	14,510
China	9,750	11,500	13,250	17,000	14,000	21,000	25,000
India	1,220	1,184	980	1,000	1,050	1,200	1,250
Japan	471	920	1,084	1,100	1,050	2,800	3,000
Total Asia	11,441	13,604	15,314	19,100	16,100	25,000	29,250
Belgium	3,298	2,840	2,825	2,850	2,400	2,650	2,650
Finland	8,170	8,580	9,100	9,200	8,800	9,000	9,000
France	280	256	305	310	400	400	400
Norway	5,021	4,927	3,939	3,800	3,500	4,000	4,100
Russia	4,748	4,759	3,587	3,200	3,300	3,500	3,500
Total Europe	21,517	21,362	19,756	19,360	18,400	19,550	19,650
Brazil	1,136	902	1,148	1,100	1,000	1,350	1,400
Canada	4,954	5,023	5,604	5,700	6,000	4,000	6,200
Total Americas	6,090	5,925	6,752	6,800	7,000	5,350	7,600
Total Oceania	3,150	3,696	3,684	4,000	3,900	3,800	6,500
DLA Deliveries	1,199	294	617	100	150	5	0
Global Total	51,846	52,358	53,630	57,140	53,090	65,775	77,510

Source: GFMS, CDI, WBMS


Cobalt Consumption 2005-2011

(tonnes)	2005	2006	2007	2008	2009	2010	2011
Total Africa/Oceania	184	186	189	191	189	192	195
China	11,000	12,200	13,900	15,250	18,500	23,000	27,000
India	600	615	635	650	700	850	1,000
Japan	13,200	13,250	13,100	12,900	11,000	12,000	12,250
South Korea	685	700	725	730	650	700	900
Taiwan	1,100	1,150	1,160	1,000	800	1,050	1,250
Thailand	240	240	250	260	260	260	260
Other Asia	1,471	1,493	1,523	1,568	1,488	1,508	1,508
Total Asia	28,296	29,648	31,293	32,358	33,398	39,368	44,168
Benelux	835	860	875	875	810	850	850
France	1,375	1,400	1,460	1,475	1,350	1,450	1,500
Germany	1,650	1,665	1,700	1,650	1,500	1,600	1,675
Italy	1,250	1,275	1,315	1,300	1,200	1,275	1,325
Russia	725	760	790	840	800	900	950
Spain	1,400	1,400	1,420	1,400	1,325	1,350	1,400
United Kingdom	2,300	2,315	2,400	2,350	2,000	2,150	2,200
Other Europe	1,913	1,935	1,935	1,945	1,759	1,828	1,828
Total Europe	11,448	11,610	11,895	11,835	10,744	11,403	11,728
United States	8,350	8,500	8,750	8,400	8,000	8,400	8,600
Other America	1,805	1,849	1,885	1,865	1,790	1,820	1,870
Total Americas	10,155	10,349	10,635	10,265	9,790	10,220	10,470
Global Total	50,892	52,602	54,817	55,454	54,921	61,983	67,361

Source: GFMS, CDI, WBMS